**MAHATMA GANDHI UNIVERSITY, KOTTAYAM**
**MGU-UGP (HONOURS) Regular EXAMINATION October 2025**
**Third SEMESTER**
**Discipline Specific Core Course (DSC)- MG3DSCNBC200- NUTRITIONAL BIOCHEMISTRY**
**(2024 ADMISSION ONWARDS)**

Duration: 2 Hours Maximum Marks: 50

## Part A
**Answer all questions**
**(Each question carries 1 mark)**

1. B- Formation of glucose from non-carbohydrate sources
2. C- Methyl group
3. B-One glycerol molecule and three fatty acids
4. B- Acetyl CoA
5. C- Liver
6. B- Phosphogluconate pathway

## Part B
**Answer any 2 questions**
**(Each question carries 2 marks )**

7. Explain the classification of deamination reactions. [U] CO2

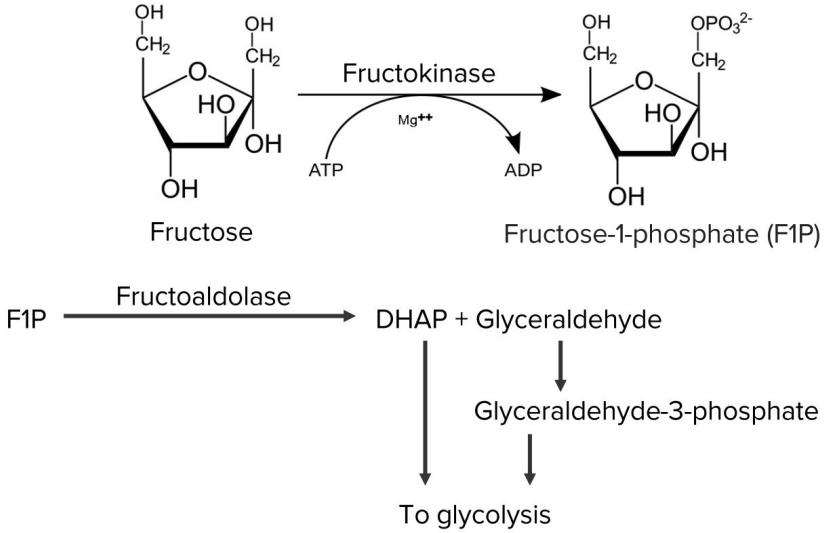
    - Oxidative deamination: removal of amino($NH_2$) group with oxidation (1 mark)
    - Non- oxidative deamination: removal of amino($NH_2$) group without oxidation,often through elimination or hydrolytic reactions. (1 mark)

8. Describe the role of thromboxanes in blood clotting and their clinical significance[U]/ (CO3)

Thromboxanes:

- Synthesized in platelets (thrombocytes) and contain a 6-membered ether ring. (1Mark)
- Promote blood clotting and reduce blood flow to the site of the clot.(1Mark)

9. Explain the metabolism of fructose [u] / [CO1]
   (each step carries ½ mark )



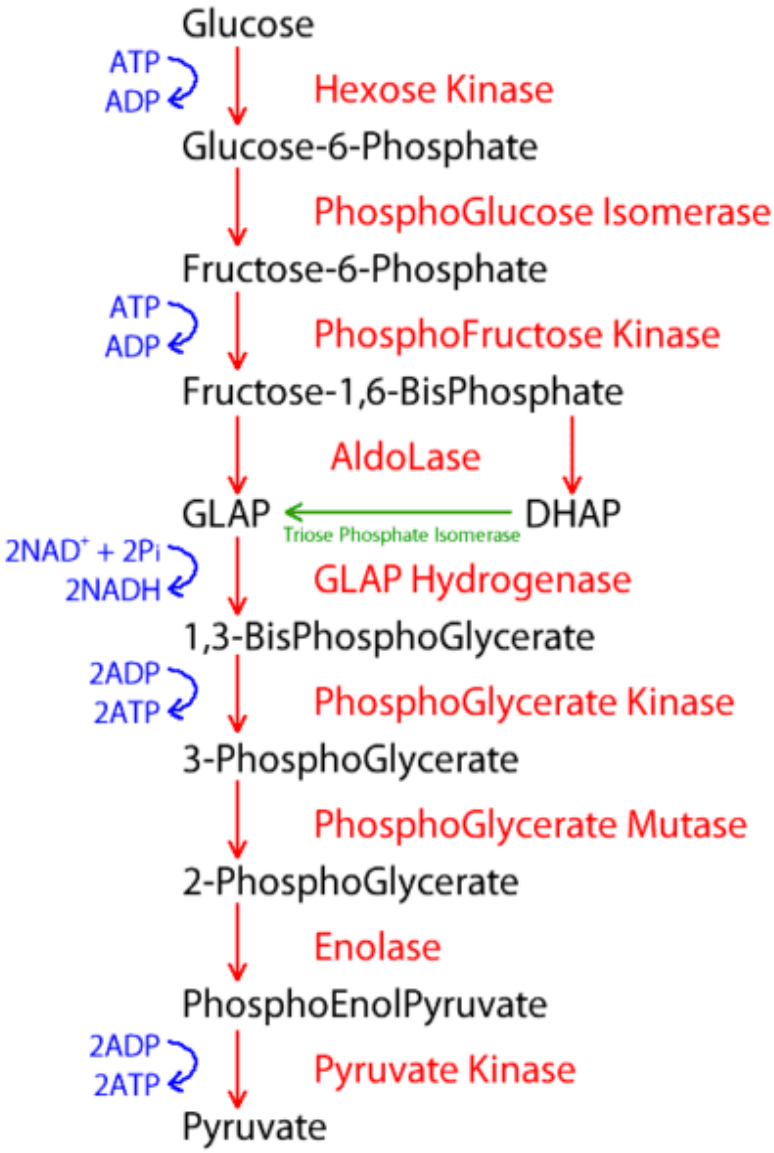

10. Define Lactic acid cycle [K] CO1

- Breakdown of pyruvate to form lactate (1 mark)
- Absence of oxygen (1 mark)

**Part C**
**Answer any 4 questions**
**Each question carries 5 marks**

11. Define Glycolysis [K] CO1

Energy requiring phase (5 marks)
Energy yielding phase (5 marks)




12. Explain any three important reactions in amino acid metabolism [U], CO2
    - Decarboxylation
    - Transmethylation
    - Deamination
    - Transamination
    - Transdeamination

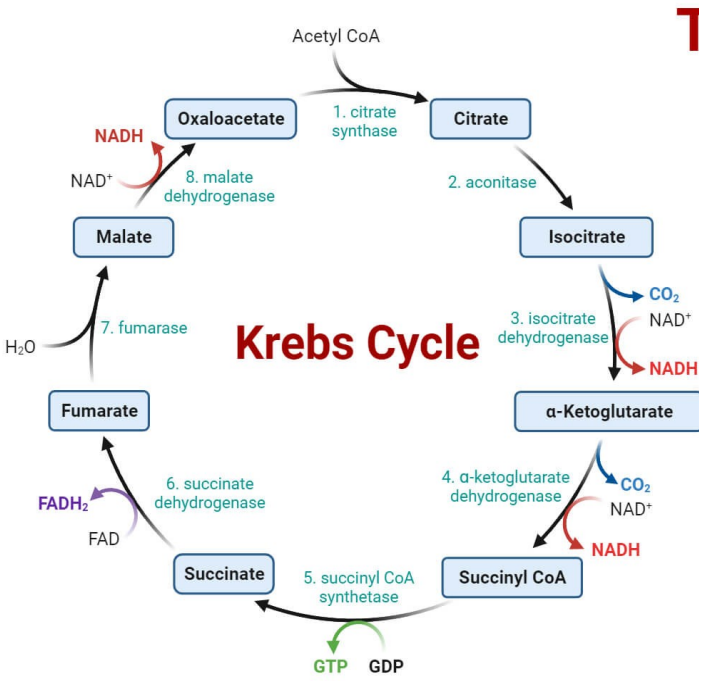
    (1 mark each for any 3 reactions
    2 marks for examples)

13. Define B-oxidation of unsaturated fatty acids and list the additional enzymes required. [K]/ (CO3)

- Normal β-oxidation continues until a double bond is reached. (1Mark)
- Enoyl-CoA isomerase converts the cis-$\Delta^3$ bond to trans-$\Delta^2$, allowing oxidation to proceed. (2Mark)
- When multiple double bonds occur, 2,4-dienoyl-CoA reductase and enoyl-CoA isomerase work together to form a usable intermediate.(2Mark)

- After these modifications, β-oxidation resumes normally to produce energy.

14. Describe Tricarboxylic acid cycle
(Each step carries ½ mark )




15. Describe how adipose tissue metabolism contributes to obesity. [U]/ (CO3)
- Excess calorie intake causes adipose tissue to store unlimited fat, leading to obesity and insulin resistance. (1Mark)
- Overloaded fat cells (adipocytes) get damaged, releasing triacylglycerol into the extracellular space.(2Mark)
- This leaked fat becomes non-functional “dead bulk”, adding weight without any metabolic role. (2 Mark)

16. Describe the energetics of glycolysis with reactions were ATP is consumed and produced. [U] CO1

Energy consumed:
Glucose ⟶ glucose -6 phosphate
Fructose-6-phosphate ⟶ Fructose 1,6 – phosphate (2 marks)

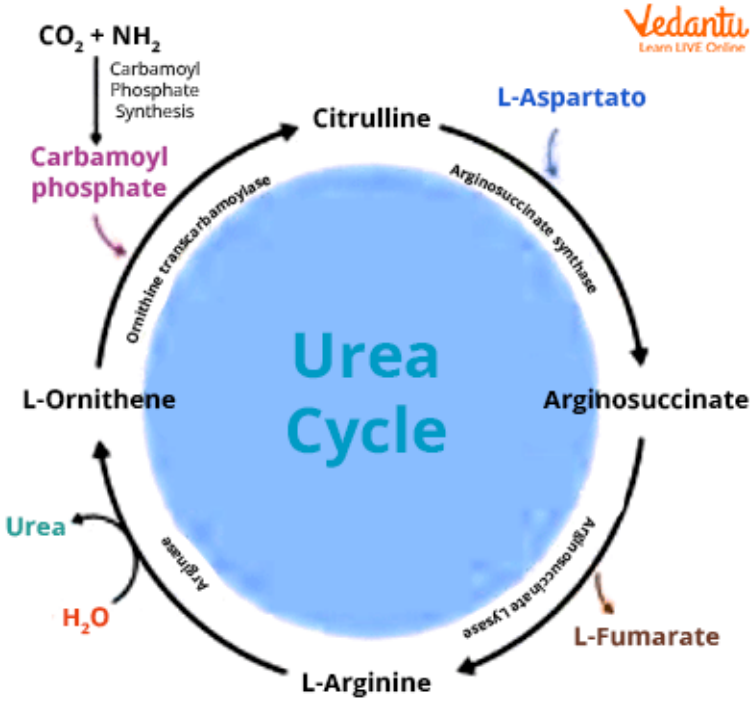
2 ATP
Energy released:
GLAP ⟶ 1,3 Bisphospho glycerate
1,3 Bisphospho glycerate ⟶ 3- phosphoglycerate
Phosphoenolpyuvate ⟶ pyruvate (3 marks)
10 ATP

Net ATP = 10-2= 8 ATP

**Part D**
**Answer any 2 questions**
**Each question carries 10 marks**

17. Describe ornithine cycle. [K] CO2

Cycle- 6 marks
Significance of urea cycle 4 marks

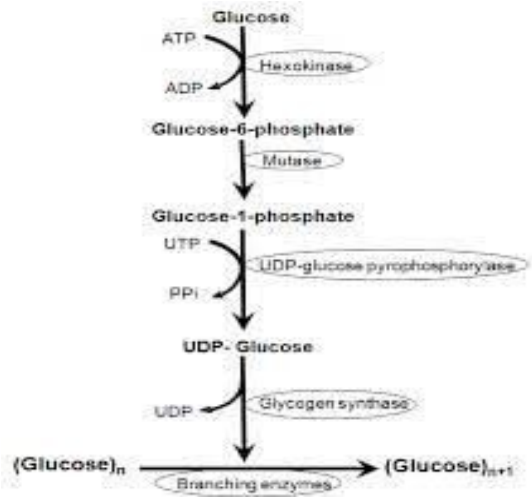



18. Discuss the role of cholesterol catabolism in the formation of bile acids and bile salts. [K]/ (CO3)

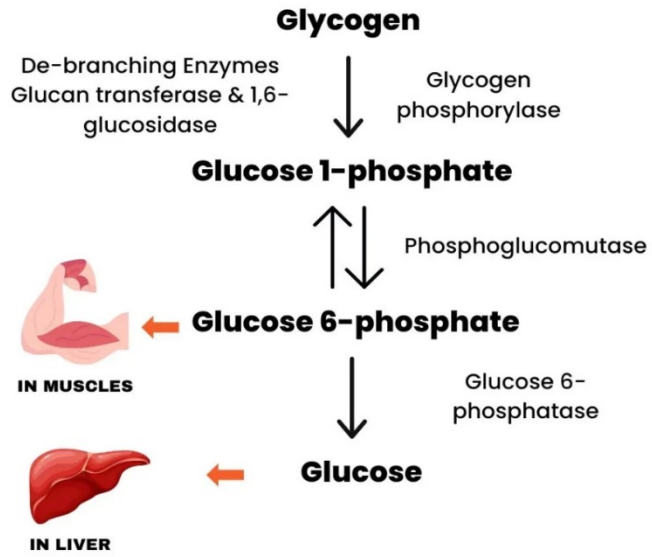
- Bile acids are the main end products of cholesterol breakdown in the liver and help remove cholesterol.(2Mark)
- The enzyme 7α-hydroxylase starts the process, forming cholic acid and chenodeoxycholic acid (primary bile acids).(2Mark)
- These acids join with glycine or taurine to form bile salts that help digest fats.(2Mark)
- Bile salts are reused through enterohepatic circulation after being reabsorbed from the intestine.(2Mark)
- Lack of bile salts can cause gallstones, and chenodeoxycholic acid may be used to dissolve them. (2Mark)

19. Describe Glycogenesis and Glycogenolysis [K], CO1

(5 marks each for each cycle)




**STEPS OF GLYCOGENOLYSIS**




20. Explain the beta oxidation of saturated fatty acid. [K]/ (CO3)

- Beta-oxidation breaks down fatty acids in the mitochondria to form acetyl-CoA, NADH, and $FADH_2$.(2Mark)
- Occurs in three main stages (8Mark)

1.Activation of Fatty Acids: Fatty acids combine with CoA using ATP and thiokinase to form fatty acyl-CoA.

2.Transport into Mitochondria: Long-chain fatty acyl-CoA uses the carnitine shuttle (carnitine acyltransferase I & II and translocase) to enter the mitochondrial matrix.

3.Steps of Beta-Oxidation Cycle:

a. Oxidation: Acyl-CoA is oxidized by acyl-CoA dehydrogenase to form trans-enoyl-CoA, producing $FADH_2$.
b. Hydration: Enoyl-CoA hydratase adds water to form β-hydroxyacyl-CoA.
c. Oxidation: β-hydroxyacyl-CoA dehydrogenase oxidizes it to β-ketoacyl-CoA, producing NADH.
d. Cleavage: Thiolase splits β-ketoacyl-CoA to release acetyl-CoA and a shortened acyl-CoA.